

Statistics & Probability



Measures of center

- 1 Find the mean of 12, 15, 18, 9, 21.
 - 2 Find the median of 4, 9, 2, 11, 7, 15.
 - 3 Find the mode of 3, 7, 5, 7, 2, 7, 9.
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Measures of spread

- 4 Find the range of the data set 22, 18, 35, 9, 27.
 - 5 Find the interquartile range (IQR) of 4, 7, 9, 12, 15, 18, 21.
 - 6 A data set has a range of 40 and a minimum value of 15. Find the maximum value.
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Basic probability

- 7 A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.
 - 8 A fair six-sided die is rolled once. Find $P(\text{even number})$.
 - 9 A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than 5})$.
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Combined probability: AND / OR

- 10 A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.
 - 11 Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
 - 12 A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.
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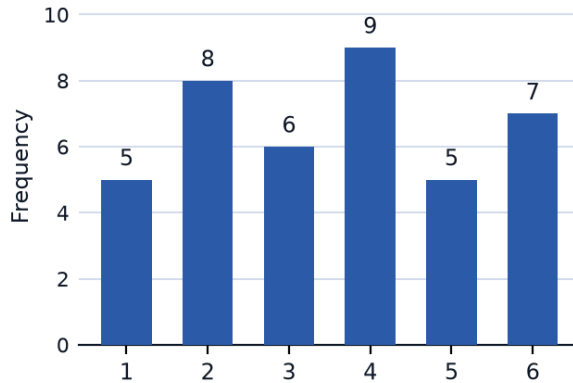
Probability without replacement

- 13 A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

- 14 A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.
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Data displays

- 15 The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.



Expected value

- 16 A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.
- 17 A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.
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Effects of transformations on statistics

- 18 A data set has mean 20. If every value in the set is increased by 5, find the new mean.
- 19 A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.
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A well-designed word problem: a survey of students

- 20** This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.
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Multiple choice — easy

- 21** Find the mean of 6, 50, 46, 36, 7.
- a. 29
 - b. 31
 - c. 145
 - d. 27
- 22** Find the mean of 41, 43, 15, 15, 6.
- a. 24
 - b. 22
 - c. 26
 - d. 120
- 23** Find the mean of 21, 18, 14, 35, 2.
- a. 90
 - b. 16
 - c. 18
 - d. 20

24 Find the mean of 48, 28, 35, 30, 19.

- a. 34
- b. 160
- c. 30
- d. 32

25 Find the mean of 23, 12, 44, 5, 26.

- a. 24
- b. 20
- c. 110
- d. 22

26 Find the mean of 26, 25, 26, 25, 48.

- a. 28
- b. 32
- c. 30
- d. 150

27 Find the mean of 4, 31, 3, 26, 1.

- a. 13
- b. 11
- c. 65
- d. 15

28 Find the mean of 22, 6, 29, 26, 42.

a. 125

b. 25

c. 23

d. 27

29 Find the mean of 45, 35, 33, 19, 13.

a. 27

b. 145

c. 31

d. 29

30 Find the mean of 9, 21, 19, 19, 22.

a. 18

b. 16

c. 20

d. 90

31 Find the mean of 6, 21, 21, 28, 29.

a. 19

b. 105

c. 21

d. 23

32 Find the mean of 50, 23, 27, 33, 12.

- a. 145
- b. 31
- c. 29
- d. 27

33 Find the mean of 3, 38, 13, 31, 5.

- a. 20
- b. 18
- c. 16
- d. 90

34 Find the mean of 14, 29, 33, 44, 30.

- a. 28
- b. 32
- c. 30
- d. 150

35 Find the mean of 31, 26, 30, 21, 47.

- a. 31
- b. 29
- c. 155
- d. 33

Multiple choice — medium

36 A jar has 11 yellow, 6 red, and 6 green marbles. Find $P(\text{red})$.

a. $\frac{6}{23}$

b. $\frac{8}{23}$

c. $\frac{11}{23}$

d. $\frac{9}{23}$

37 A jar has 3 green, 5 blue, and 12 purple marbles. Find $P(\text{blue})$.

a. $\frac{3}{20}$

b. $\frac{7}{20}$

c. $\frac{3}{5}$

d. $\frac{1}{4}$

38 A jar has 6 red, 2 orange, and 2 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{5}$

b. $\frac{3}{5}$

c. $\frac{4}{5}$

d. $\frac{9}{10}$

39 A jar has 4 red, 8 purple, and 4 green marbles. Find $P(\text{red})$.

a. $\frac{7}{16}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

40 A jar has 2 green, 1 orange, and 10 yellow marbles. Find $P(\text{orange})$.

a. $\frac{1}{13}$

b. $\frac{2}{13}$

c. $\frac{3}{13}$

d. $\frac{10}{13}$

41 A jar has 9 red, 2 green, and 10 blue marbles. Find $P(\text{red})$.

a. $\frac{2}{21}$

b. $\frac{10}{21}$

c. $\frac{11}{21}$

d. $\frac{3}{7}$

42 A jar has 1 yellow, 5 purple, and 5 orange marbles. Find $P(\text{yellow})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$

c. $\frac{3}{11}$

d. $\frac{1}{11}$

43 A jar has 2 red, 5 purple, and 1 orange marbles. Find $P(\text{red})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$

44 A jar has 7 purple, 11 orange, and 10 green marbles. Find $P(\text{green})$.

a. $\frac{11}{28}$

b. $\frac{3}{7}$

c. $\frac{1}{4}$

d. $\frac{5}{14}$

45 A jar has 7 yellow, 1 green, and 9 red marbles. Find $P(\text{yellow})$.

a. $\frac{7}{17}$

b. $\frac{9}{17}$

c. $\frac{1}{17}$

d. $\frac{10}{17}$

46 A jar has 10 red, 8 purple, and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{3}$

b. $\frac{8}{27}$

c. $\frac{10}{27}$

d. $\frac{4}{9}$

47 A jar has 5 yellow, 9 orange, and 3 blue marbles. Find $P(\text{orange})$.

a. $\frac{9}{17}$

b. $\frac{5}{17}$

c. $\frac{11}{17}$

d. $\frac{3}{17}$

48 A jar has 12 green, 12 orange, and 2 blue marbles. Find $P(\text{blue})$.

a. $\frac{5}{26}$

b. $\frac{1}{13}$

c. $\frac{2}{13}$

d. $\frac{6}{13}$

49 A jar has 7 red, 1 yellow, and 9 green marbles. Find $P(\text{red})$.

a. $\frac{10}{17}$

b. $\frac{7}{17}$

c. $\frac{9}{17}$

d. $\frac{1}{17}$

50 A jar has 9 blue, 10 orange, and 3 red marbles. Find $P(\text{blue})$.

a. $\frac{5}{11}$

b. $\frac{9}{22}$

c. $\frac{3}{22}$

d. $\frac{1}{2}$

Multiple choice — hard

51 Two independent events have $P(A) = 0.1$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

a. 0.28

b. 0.13

c. 0.18

d. 0.08

- 52 Two independent events have $P(A) = 0.1$ and $P(B) = 0.7$. Find $P(A \text{ and } B)$.
- a. 0.27
 - b. 0.17
 - c. 0.12
 - d. 0.07
- 53 Two independent events have $P(A) = 0.8$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.
- a. 0.32
 - b. 0.42
 - c. 0.52
 - d. 0.22
- 54 Two independent events have $P(A) = 0.5$ and $P(B) = 0.6$. Find $P(A \text{ and } B)$.
- a. 0.40
 - b. 0.20
 - c. 0.50
 - d. 0.30
- 55 Two independent events have $P(A) = 0.8$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
- a. 0.40
 - b. 0.30
 - c. 0.50
 - d. 0.60

- 56 A bag has 3 red and 10 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.
- a. $\frac{3}{13}$
 - b. $\frac{15}{26}$
 - c. $\frac{1}{26}$
 - d. $\frac{5}{13}$
- 57 A bag has 7 red and 9 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.
- a. $\frac{7}{16}$
 - b. $\frac{21}{40}$
 - c. $\frac{3}{10}$
 - d. $\frac{7}{40}$
- 58 A bag has 6 red and 6 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.
- a. $\frac{5}{22}$
 - b. $\frac{1}{2}$
 - c. $\frac{6}{11}$
 - d. $\frac{31}{132}$
- 59 A bag has 7 red and 4 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.
- a. $\frac{21}{55}$
 - b. $\frac{28}{55}$
 - c. $\frac{7}{11}$
 - d. $\frac{6}{55}$

- 60 A bag has 5 red and 3 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.
- a. $\frac{3}{28}$
 - b. $\frac{5}{14}$
 - c. $\frac{15}{28}$
 - d. $\frac{5}{8}$
- 61 A spinner has 4 equal sections labeled 15, 20, 35, and 45 points. Find the expected value of one spin.
- a. 28.75
 - b. 29.75
 - c. 27.75
 - d. 31.75
- 62 A spinner has 4 equal sections labeled 10, 20, 25, and 40 points. Find the expected value of one spin.
- a. 26.75
 - b. 24.75
 - c. 23.75
 - d. 22.75
- 63 A spinner has 4 equal sections labeled 5, 25, 30, and 35 points. Find the expected value of one spin.
- a. 26.75
 - b. 22.75
 - c. 24.75
 - d. 23.75

- 64** A spinner has 4 equal sections labeled 15, 20, 30, and 40 points. Find the expected value of one spin.
- a. 27.25
 - b. 26.25
 - c. 29.25
 - d. 25.25
- 65** A spinner has 4 equal sections labeled 5, 15, 35, and 45 points. Find the expected value of one spin.
- a. 28
 - b. 24
 - c. 26
 - d. 25